

Norm Compression for Scaled-Unitary Blocks

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Status: candidate partial result, likely valid. Audenaert's general norm-compression conjecture remains open in this packet. The theorem below proves all finite Schatten exponents for arbitrary collections of common-size blocks that are scalar multiples of unitaries.

1 Source conjecture

Koenraad M. R. Audenaert's paper *On a Norm Compression Inequality for $2 \times N$ Partitioned Block Matrices*, arXiv:math/0702186, formulates Conjecture 1 on PDF page 2 [1]. For

$$T = \begin{pmatrix} A_1 & A_2 & \cdots & A_N \\ B_1 & B_2 & \cdots & B_N \end{pmatrix}, \quad \mathcal{C}_p(T) = \begin{pmatrix} \|A_1\|_p & \cdots & \|A_N\|_p \\ \|B_1\|_p & \cdots & \|B_N\|_p \end{pmatrix},$$

it asks whether

$$\|T\|_p \geq \|\mathcal{C}_p(T)\|_p, \quad 1 \leq p \leq 2, \quad (1)$$

$$\|T\|_p \leq \|\mathcal{C}_p(T)\|_p, \quad p \geq 2. \quad (2)$$

Here every norm is the Schatten norm at the displayed exponent.

In this paper, I propose a norm compression inequality, based on the Schatten norms, for $2 \times N$ block-partitioned matrices:

$$T = \begin{pmatrix} A_1 & A_2 & \cdots & A_N \\ B_1 & B_2 & \cdots & B_N \end{pmatrix}.$$

I denote by $\mathcal{C}_p(T)$ its Schatten p -norm compression,

$$\mathcal{C}_p(T) = \begin{pmatrix} \|A_1\|_p & \|A_2\|_p & \cdots & \|A_N\|_p \\ \|B_1\|_p & \|B_2\|_p & \cdots & \|B_N\|_p \end{pmatrix}.$$

I conjecture the following:

Conjecture 1 *Let T be a general matrix partitioned in $2 \times N$ blocks, and let $\mathcal{C}_p(T)$ be its norm compression using the Schatten p -norm, then the following norm compression inequalities hold:*

$$\|T\|_p \geq \|\mathcal{C}_p(T)\|_p, \quad 1 \leq p \leq 2 \tag{1}$$

$$\|T\|_p \leq \|\mathcal{C}_p(T)\|_p, \quad p \geq 2. \tag{2}$$

Figure 1: The definition and Conjecture 1 as rendered from page 2 of the source paper.

2 Definitions and scope

For a finite matrix X , its Schatten p -norm is

$$\|X\|_p = (\text{Tr } |X|^p)^{1/p}, \quad 1 \leq p < \infty.$$

We treat the subcase in which all blocks are $d \times d$ and

$$A_i = a_i U_i, \quad B_i = b_i V_i, \quad a_i, b_i \geq 0, \quad U_i, V_i \in \text{U}(d).$$

The unitaries may depend arbitrarily on i . Thus the blocks may be full rank, non-proportional, and noncommuting. This differs from the rank-one and proportional-block families handled in the source.

3 Proof intuition

Scaled unitaries make the two diagonal blocks of TT^* scalar. All remaining geometry is contained in one cross term

$$Z = \sum_i a_i b_i U_i V_i^*.$$

After taking a singular-value decomposition of Z , the $2d \times 2d$ matrix TT^* splits into d scalar 2×2 problems. If s_j is a singular value of Z , the triangle inequality gives $s_j \leq c := \sum_i a_i b_i$. The scalar compression is exactly the perfectly aligned endpoint $s = c$, repeated d times.

The sum of the q th powers of the two eigenvalues of $\begin{pmatrix} \alpha & s \\ s & \beta \end{pmatrix}$ is decreasing in s for $q \leq 1$ and increasing for $q \geq 1$. Substituting $q = p/2$ produces the desired reversal at $p = 2$. The unrestricted conjecture remains open because general blocks do not make the diagonal blocks of TT^* scalar, so this singular-value decoupling is unavailable.

4 The partial theorem

Theorem 1 (Scaled-unitary norm compression). *Let $N, d \geq 1$, let $a_i, b_i \geq 0$, and let $U_i, V_i \in \text{U}(d)$ for $1 \leq i \leq N$. Form*

$$T = \begin{pmatrix} a_1 U_1 & \cdots & a_N U_N \\ b_1 V_1 & \cdots & b_N V_N \end{pmatrix}.$$

Then Audenaert's norm-compression inequalities (1) and (2) hold for every $1 \leq p < \infty$.

Proof. Fix p and write $q = p/2$. Set

$$\alpha = \sum_{i=1}^N a_i^2, \quad \beta = \sum_{i=1}^N b_i^2, \quad c = \sum_{i=1}^N a_i b_i, \quad Z = \sum_{i=1}^N a_i b_i U_i V_i^*.$$

The unitary hypotheses give

$$TT^* = \begin{pmatrix} \alpha I_d & Z \\ Z^* & \beta I_d \end{pmatrix}. \tag{3}$$

Let $s_1, \dots, s_d$ be the singular values of Z . If $Z = L \text{diag}(s_1, \dots, s_d) R^*$ is a singular-value decomposition, conjugating (3) by $\text{diag}(L, R)$ and permuting coordinates shows that TT^* is unitarily equivalent to the direct sum

$$\bigoplus_{j=1}^d \begin{pmatrix} \alpha & s_j \\ s_j & \beta \end{pmatrix}.$$

Moreover,

$$0 \leq s_j \leq \|Z\|_\infty \leq \sum_{i=1}^N a_i b_i \|U_i V_i^*\|_\infty = c. \quad (4)$$

For $0 \leq s \leq c$, define

$$\lambda_\pm(s) = \frac{\alpha + \beta \pm \sqrt{(\alpha - \beta)^2 + 4s^2}}{2}, \quad F_q(s) = \lambda_+(s)^q + \lambda_-(s)^q.$$

Cauchy–Schwarz gives $c^2 \leq \alpha\beta$, so these are the two nonnegative eigenvalues throughout the required interval. The preceding unitary equivalence yields

$$\|T\|_p^p = \text{Tr}((TT^*)^q) = \sum_{j=1}^d F_q(s_j). \quad (5)$$

For $s > 0$ at points where both eigenvalues are positive, direct differentiation gives

$$F'_q(s) = \frac{2qs}{\sqrt{(\alpha - \beta)^2 + 4s^2}} (\lambda_+(s)^{q-1} - \lambda_-(s)^{q-1}). \quad (6)$$

Thus F_q is nonincreasing when $0 < q \leq 1$ and nondecreasing when $q \geq 1$. The same conclusion at $s = 0$ and when $\lambda_-(s) = 0$ follows by continuity. At $q = 1$, $F_1(s) = \alpha + \beta$ is constant.

It remains to identify the endpoint $s = c$ with the norm compression. Since the singular values of each unitary equal one,

$$\|a_i U_i\|_p = d^{1/p} a_i, \quad \|b_i V_i\|_p = d^{1/p} b_i.$$

Consequently

$$\mathcal{C}_p(T) \mathcal{C}_p(T)^* = d^{2/p} \begin{pmatrix} \alpha & c \\ c & \beta \end{pmatrix},$$

and, because $2q/p = 1$,

$$\|\mathcal{C}_p(T)\|_p^p = \text{Tr}((\mathcal{C}_p(T) \mathcal{C}_p(T)^*)^q) = d F_q(c). \quad (7)$$

If $1 \leq p \leq 2$, then $q \leq 1$; equations (4)–(7) give

$$\|T\|_p^p = \sum_j F_q(s_j) \geq d F_q(c) = \|\mathcal{C}_p(T)\|_p^p.$$

If $p \geq 2$, the monotonicity reverses and gives the opposite inequality. Taking p th roots proves the theorem. $\square$

Remark 1. *At $p = 2$ there is equality for every choice of blocks. Away from $p = 2$, the proof also shows equality whenever every singular value of Z equals c ; for example, this occurs when all relative unitaries $U_i V_i^*$ agree.*

5 Computational verification

The reusable script `code/verify_scaled_unitary.py` generates seeded complex Haar unitaries and compares direct Schatten norms with both formulas (5) and (7). It also checks the inequality direction for $p \in \{1, 1.2, 1.5, 1.9, 2, 2.2, 2.8, 3.5, 5, 8\}$ across 5,000 random instances with $1 \leq d \leq 6$ and $2 \leq N \leq 7$. The exact command and numerical residuals appear in `verification.md`. These tests are not used as proof.

6 Limitations and bounded novelty check

The theorem does not handle arbitrary blocks, blocks with non-flat singular values, or the full matrix Hanner conjecture. It is therefore promoted only as a partial result.

The run’s cheap indexes were searched by arXiv id, title, and the phrases “norm compression”, “Schatten”, and “Hanner”. A bounded web/arXiv search combined the source title and author with “unitary blocks”, “scaled unitaries”, and “norm compression unitary”. It recovered the source and nearby PSD/diagonal results, but no exact statement of Theorem 1. This is not an exhaustive novelty claim.

7 Human-review recommendation

The candidate is suitable for mathematical review. The most useful checks are: (i) the unitary equivalence following (3); (ii) the monotonicity sign in (6); and (iii) the factor d in (7). Until review, retain the status *candidate partial result, likely valid*.

8 Reference

References

- [1] K. M. R. Audenaert, *On a Norm Compression Inequality for $2 \times N$ Partitioned Block Matrices*, Linear Algebra Appl. 428 (2008), 781–795; arXiv:math/0702186.